

MCP

Reference

#	Tool	Category	Read-Only
1	create_project	Project Management	false
2	get_project	Project Management	true
3	list_projects	Project Management	true
4	list_screens	Screen Management	true
5	get_screen	Screen Management	true
6	generate_screen_from_text	AI Generation	false
7	edit_screens	AI Generation	false
8	generate_variants	AI Generation	false
9	create_design_system	Design Systems	false
10	update_design_system	Design Systems	false
11	list_design_systems	Design Systems	true
12	apply_design_system	Design Systems	false

Tools

create_project

Creates a new Stitch project. A project is a container for UI designs and frontend code.

ANNOTATIONS

destructive

Input

```
{
  "type": "object",
  "properties": {
    "title": {
      "type": "string",
      "description": "Optional. The title of the project."
    }
  },
  "required": []
}
```

Output

Returns the created Project resource with generated name, title, and metadata.

Example Prompts

PROMPT

Create a new Stitch project called 'Marketing Site'
Start a new design project
Set up a new project for my app

get_project

Retrieves the details of a specific Stitch project using its project name.

ANNOTATIONS

readOnly

Input

```
{
  "type": "object",
  "properties": {
    "name": {
      "type": "string",
      "description": "Required. Resource name. Format: `projects/{project}`. Example: `projects/4044680601076201931`"
    }
  },
  "required": ["name"]
}
```

Output

Returns a Project resource object.

Example Prompts

PROMPT

Get the details of project 4044680601076201931
Show me the project info
What's in my project?

list_projects

Lists all Stitch projects accessible to the user. By default, it lists projects owned by the user.

ANNOTATIONS

readOnly

Input

```
{
  "type": "object",
  "properties": {
    "filter": {
      "type": "string",
      "description": "Optional. AIP-160 filter on `view` field. Supported: `view=owned` (default), `view=shared`."
    }
  },
  "required": []
}
```

Output

Array of Project resource objects.

Example Prompts

PROMPT

List all my Stitch projects
Show me my projects
List shared projects

list_screens

Lists all screens within a given Stitch project.

ANNOTATIONS

readOnly

Input

```
{
  "type": "object",
  "properties": {
    "projectId": {
      "type": "string",
      "description": "Required. Project ID (e.g., '4044680601076201931'), without `projects/` prefix."
    }
  },
  "required": ["projectId"]
}
```

Output

Array of [Screen](#) objects.

Example Prompts

PROMPT

List all screens in project 12345
Show me the designs in this project
What screens are in my project?

get_screen

Retrieves the details of a specific screen within a project.

ANNOTATIONS

readOnly

Input

```
{
  "type": "object",
  "properties": {
    "name": {
      "type": "string",
      "description": "Required. Resource name. Format:
`projects/{project}/screens/{screen}`"
    },
    "projectId": {
      "type": "string",
      "description": "Required (deprecated). Project ID without
prefix."
    },
    "screenId": {
      "type": "string",
      "description": "Required (deprecated). Screen ID without
prefix."
    }
  },
  "required": ["name", "projectId", "screenId"]
}
```

projectId and screenId are deprecated in favor of name (resource name format), but all three are currently required.

Output

Returns a [Screen](#) object with htmlCode, screenshot, figmaExport file references and download URLs.

Example Prompts

PROMPT

Get the details of screen abc123 in project 12345
Show me the screen info
Check the status of this screen

generate_screen_from_text

Generates a new screen from a text prompt. This process normally takes **a few minutes**. Avoid retrying even when you encounter connection errors. Connection errors don't necessarily mean failure, try using `get_screen` after a few minutes. If `output_components` contains suggestions, present them to the user. If accepted, call again with the suggestion as the new prompt.

ANNOTATIONS

destructive

Input


```
{
  "type": "object",
  "properties": {
    "projectId": {
      "type": "string",
      "description": "Required. Project ID without prefix."
    },
    "prompt": {
      "type": "string",
      "description": "Required. Text prompt describing the screen to generate."
    },
    "deviceType": {
      "type": "string",
      "enum": ["DEVICE_TYPE_UNSPECIFIED", "MOBILE", "DESKTOP", "TABLET", "AGNOSTIC"]
    },
    "modelId": {
      "type": "string",
      "enum": ["MODEL_ID_UNSPECIFIED", "GEMINI_3_PRO", "GEMINI_3_FLASH", "GEMINI_3_1_PRO"]
    }
  },
  "required": ["projectId", "prompt"]
}
```

GEMINI_3_PRO is deprecated. Use GEMINI_3_1_PRO or GEMINI_3_FLASH instead.

Output

Returns generated [Screen](#) objects and [SessionOutputComponent](#) entries (text, suggestions, or design references).

Example Prompts

PROMPT

Generate a mobile login screen for project 12345
Design a dashboard page with charts and statistics
Create a landing page with a hero section, features grid, and footer
Build a settings page for a dark-themed app

edit_screens

Edits existing screens using a text prompt. Similar to `generate_screen_from_text`, this process normally takes **a few minutes**. Avoid retrying even when you encounter connection errors. Connection errors don't necessarily mean failure, try using `get_screen` after a few minutes.

ANNOTATIONS

destructive

Input

```
{
  "type": "object",
  "properties": {
    "projectId": {
      "type": "string",
      "description": "Required. Project ID without prefix."
    },
    "selectedScreenIds": {
      "type": "array",
      "items": { "type": "string" },
      "description": "Required. Screen IDs to edit, without `screens/` prefix."
    },
    "prompt": {
      "type": "string",
      "description": "Required. Edit instruction."
    },
    "deviceType": {
      "type": "string",
      "enum": ["DEVICE_TYPE_UNSPECIFIED", "MOBILE", "DESKTOP", "TABLET", "AGNOSTIC"]
    },
    "modelId": {
      "type": "string",
      "enum": ["MODEL_ID_UNSPECIFIED", "GEMINI_3_PRO", "GEMINI_3_FLASH", "GEMINI_3_1_PRO"]
    }
  },
  "required": ["projectId", "selectedScreenIds", "prompt"]
}
```

GEMINI_3_PRO is deprecated. Use GEMINI_3_1_PRO or GEMINI_3_FLASH instead.

Output

Returns the updated [Screen](#) objects.

Example Prompts

PROMPT



Change the button color to blue on screen abc123
Make the hero text larger and add a subtitle
Add a navigation bar to these screens
Switch the layout to a two-column grid

generate_variants

Generates design variants of existing screens.

ANNOTATIONS

destructive

Input

```
{
  "type": "object",
  "properties": {
    "projectId": {
      "type": "string",
      "description": "Required. Project ID without prefix."
    },
    "selectedScreenIds": {
      "type": "array",
      "items": { "type": "string" },
      "description": "Required. Screen IDs to generate variants for."
    },
    "prompt": {
      "type": "string",
      "description": "Required. Text guiding variant generation."
    },
    "variantOptions": {
      "$ref": "#/$defs/VariantOptions",
      "description": "Required. Variant configuration."
    },
    "deviceType": {
      "type": "string",
      "enum": ["DEVICE_TYPE_UNSPECIFIED", "MOBILE", "DESKTOP",
"TABLET", "AGNOSTIC"]
    },
    "modelId": {
      "type": "string",
      "enum": ["MODEL_ID_UNSPECIFIED", "GEMINI_3_PRO",
"GEMINI_3_FLASH", "GEMINI_3_1_PRO"]
    }
  },
  "required": ["projectId", "selectedScreenIds", "prompt",
"variantOptions"]
}
```

GEMINI_3_PRO is deprecated. Use GEMINI_3_1_PRO or GEMINI_3_FLASH instead.

VariantOptions

```
{
  "type": "object",
  "properties": {
    "variantCount": {
      "type": "integer",
      "description": "Number of variants (1-5). Default: 3."
    },
    "creativeRange": {
      "type": "string",
      "enum": ["CREATIVE_RANGE_UNSPECIFIED", "REFINE", "EXPLORE",
"REIMAGINE"]
    },
    "aspects": {
      "type": "array",
      "items": {
        "type": "string",
        "enum": ["VARIANT_ASPECT_UNSPECIFIED", "LAYOUT",
"COLOR_SCHEME", "IMAGES", "TEXT_FONT", "TEXT_CONTENT"]
      }
    }
  }
}
```

Output

Returns generated variant [Screen](#) objects.

Example Prompts

PROMPT

Generate 3 variants of screen abc123 focusing on layout and color
Reimagine my landing page with radical variations
Create subtle refinements of these screens
Show me 5 different color schemes for this design

create_design_system

Creates a new design system for a project. Establishes foundational design tokens (colors, typography, shapes, appearance) that apply across all screens.

ANNOTATIONS

destructive

Input

```
{
  "type": "object",
  "properties": {
    "designSystem": {
      "$ref": "#/$defs/DesignSystem",
      "description": "Required. The design system to create."
    },
    "projectId": {
      "type": "string",
      "description": "Optional. Project ID. If empty, creates a global (unassociated) asset."
    }
  },
  "required": ["designSystem"]
}
```

DesignSystem

```
{
  "type": "object",
  "properties": {
    "displayName": {
      "type": "string",
      "description": "Required. Display name."
    },
    "theme": {
      "$ref": "#/$defs/DesignTheme",
      "description": "Required. Theme configuration."
    }
  },
  "required": ["displayName", "theme"]
}
```

DesignTheme


```
{
  "type": "object",
  "properties": {
    "colorMode": { "type": "string", "enum":
["COLOR_MODE_UNSPECIFIED", "LIGHT", "DARK"] },
    "headlineFont": { "type": "string", "enum": ["FONT_UNSPECIFIED",
"INTER", "DM_SANS", "GEIST", "..."] },
    "bodyFont": { "type": "string", "enum": ["FONT_UNSPECIFIED",
"INTER", "DM_SANS", "GEIST", "..."] },
    "labelFont": { "type": "string", "enum": ["FONT_UNSPECIFIED",
"INTER", "DM_SANS", "GEIST", "..."] },
    "roundness": { "type": "string", "enum": ["ROUNDNESS_UNSPECIFIED",
"ROUND_FOUR", "ROUND_EIGHT", "ROUND_TWELVE", "ROUND_FULL"] },
    "customColor": { "type": "string", "description": "Required. Hex
color seed for the dynamic color system." },
    "colorVariant": { "type": "string", "enum":
["COLOR_VARIANT_UNSPECIFIED", "MONOCHROME", "NEUTRAL", "TONAL_SPOT",
"VIBRANT", "EXPRESSIVE", "FIDELITY", "CONTENT", "RAINBOW",
"FRUIT_SALAD"] },
    "overridePrimaryColor": { "type": "string", "description":
"Optional. Override primary color (hex)." },
    "overrideSecondaryColor": { "type": "string", "description":
"Optional. Override secondary color (hex)." },
    "overrideTertiaryColor": { "type": "string", "description":
"Optional. Override tertiary color (hex)." },
    "overrideNeutralColor": { "type": "string", "description":
"Optional. Override neutral color (hex)." },
    "designMd": { "type": "string", "description": "Optional. Markdown
design instructions." }
  },
  "required": ["colorMode", "headlineFont", "bodyFont", "roundness",
"customColor"]
}
```

Output

Returns the created design system Asset.

Example Prompts

PROMPT



Create a dark mode design system with Inter font and round corners
Set up a design system with blue as the primary color
Create a brand identity with Geist font and minimal roundness

update_design_system

Updates an existing design system. Identifies the asset by its name field.

ANNOTATIONS

destructive

Input

```
{
  "type": "object",
  "properties": {
    "name": {
      "type": "string",
      "description": "Required. Resource name. Format: `assets/{asset}`. Example: `assets/15996705518239280238`"
    },
    "projectId": {
      "type": "string",
      "description": "Required. Project ID without prefix."
    },
    "designSystem": {
      "$ref": "#/$defs/DesignSystem",
      "description": "Required. The updated design system content."
    }
  },
  "required": ["name", "projectId", "designSystem"]
}
```

Uses the same [DesignSystem](#) and [DesignTheme](#) schemas as `create_design_system`.

Output

Returns the updated design system Asset.

Example Prompts

PROMPT

Update the design system to use dark mode
Change the font to Geist in our design system
Update the roundness to fully rounded

list_design_systems

Lists all design systems for a project (or global design systems if no projectId).

ANNOTATIONS

readOnly

Input

```
{
  "type": "object",
  "properties": {
    "projectId": {
      "type": "string",
      "description": "Optional. Project ID. If empty, lists global design systems."
    }
  },
  "required": []
}
```

Output

Array of Asset objects containing design systems.

Example Prompts

PROMPT

List all design systems in project 12345
Show me the available design systems
What design systems do I have?

apply_design_system

Applies a design system to one or more screens, modifying their appearance to match the system's tokens (colors, fonts, shapes).

ANNOTATIONS

destructive

Input

```
{
  "type": "object",
  "properties": {
    "projectId": {
      "type": "string",
      "description": "Required. Project ID without prefix."
    },
    "selectedScreenInstances": {
      "type": "array",
      "items": { "$ref": "#/$defs/SelectedScreenInstance" },
      "description": "Required. Screen instances to update (from `get_project`)."
    },
    "assetId": {
      "type": "string",
      "description": "Required. Design system asset ID (from `list_design_systems`), without `assets/` prefix."
    }
  },
  "required": ["projectId", "selectedScreenInstances", "assetId"]
}
```

SelectedScreenInstance

```
{
  "type": "object",
  "properties": {
    "id": {
      "type": "string",
      "description": "Required. The screen instance ID (NOT the source screen ID). Available from `get_project`."
    },
    "sourceScreen": {
      "type": "string",
      "description": "Required. Resource name. Format: `projects/{project}/screens/{screen}`"
    }
  },
  "required": ["id", "sourceScreen"]
}
```

Output

Returns the updated [Screen](#) objects.

Example Prompts

PROMPT



Apply the blue design system to all screens in project 12345
Restyle these screens with the brand identity
Use design system abc to update my screens

Shared Types

Screen

A generated screen design within a Stitch project.

Field	Type	Description
name	string	Resource name. Format: projects/{project}/screens/{screen}
id	string	<i>(Deprecated)</i> The screen id.
title	string	Title of the screen.
prompt	string	Prompt used to generate the screen.
screenshot	File	Screenshot image of the screen.
htmlCode	File	HTML code of the screen.
figmaExport	File	Figma export of the screen.
theme	DesignTheme	Theme used for generation.
deviceType	DeviceType	Device type.
screenType	ScreenType	Screen type.
screenMetadata	ScreenMetadata	Metadata (agent, status, display mode).
width	string	Width of the screen.
height	string	Height of the screen.
groupId	string	Group id for related screens (e.g., variants).
groupName	string	Display name of the group.
generatedBy	string	Generator identifier.
isCreatedByClient	boolean	true if created via API upload, false if by

agents.

File

Field	Type	Description
name	string	Resource name. Format: projects/{project}/files/{file}
downloadUrl	string	URL for download. FIFE base URL for images.
mimeType	string	E.g., "image/png", "text/html".
fileContentBase64	string	<i>(Write-only)</i> Base64-encoded content for uploads.
uploadBlobId	string	BlobId from upload service.
userFeedback	UserFeedback	<i>(Read-only)</i> Latest feedback.

DesignSystem

Represents visual/functional design guidelines used to generate consistent, branded designs.

Field	Type	Description
displayName	string	Display name of the design system.
theme	DesignTheme	Theme configuration.

Asset

Wrapper for design system resources (used by `update_design_system`).

Field	Type	Description
name	string	Resource name. Format: assets/{asset}
designSystem	DesignSystem	The design system content.
version	string	<i>(Read-only)</i> Version counter. 0 = unversioned.

DesignTheme

Field	Type	Required	Description
colorMode	ColorMode	true	Light or dark mode.
headlineFont	Font	true	Headline typography.
bodyFont	Font	true	Body typography.
labelFont	Font	false	Label typography.
roundness	Roundness	true	Corner roundness.
customColor	string	true	Seed color for dynamic color system (hex, e.g., "#ff0000").
colorVariant	ColorVariant	false	Dynamic color system variant.
overridePrimaryColor	string	false	Override primary color (hex).
overrideSecondaryColor	string	false	Override secondary color (hex).

overrideTertiaryColor	string	false	Override tertiary color (hex).
overrideNeutralColor	string	false	Override neutral color (hex).
designMd	string	false	Markdown design instructions.

ScreenMetadata

Field	Type	Description
agentType	AgentTypeEnum	TURBO_AGENT, PRO_AGENT, IMAGE_AGENT, GENIE_AGENT, IMAGE_PRO_AGENT, HATTER_AGENT, GEMINI_3_AGENT
status	ScreenStatusEnum	IN_PROGRESS, COMPLETE, FAILED
statusMessage	string	Human-readable status from agent.
displayMode	DisplayModeEnum	SCREENSHOT, HTML, CODE, MARKDOWN, STICKY_NOTE
isRemixed	boolean	Whether screen was created from a remix.
componentRegions	ComponentRegion[]	Component region annotations.

SessionOutputComponent

Returned from generation tools in output_components.

Field	Type	Description
text	string	Agent text response.
suggestion	string	Suggested follow-up for the user.
design	Design	Generated design reference.

UserFeedback

Field	Type	Description
rating	string	POSITIVE or NEGATIVE.
comment	string	Additional comment.
designFeedbackReason	string	DESIGN_DOESNT_MATCH_PROMPT, EDIT_DOESNT_MATCH_PROMPT, COMPONENT_ISSUE, INCORRECT_THEME, etc.

VariantOptions

Field	Type	Description
variantCount	integer	Variants to generate (1–5). Default: 3.
creativeRange	CreativeRange	Creative freedom level.
aspects	VariantAspect[]	Aspects to focus on.

SelectedScreenInstance

Used by `apply_design_system`.

Field	Type	Required	Description
<code>id</code>	<code>string</code>	<code>true</code>	Screen instance ID (from <code>get_project</code>), not the source screen ID.
<code>sourceScreen</code>	<code>string</code>	<code>true</code>	Resource name. Format: <code>projects/{project}/screens/{screen}</code>

Enums

DeviceType Enum

Value	Description
<code>DEVICE_TYPE_UNSPECIFIED</code>	Unspecified.
<code>MOBILE</code>	Mobile device.
<code>DESKTOP</code>	Desktop device.
<code>TABLET</code>	Tablet device.
<code>AGNOSTIC</code>	Not tied to a device.

ModelId Enum

Value	Description
<code>MODEL_ID_UNSPECIFIED</code>	Unspecified.

GEMINI_3_PRO	(<i>Deprecated</i>) Gemini 3 Pro. Use GEMINI_3_1_PRO or GEMINI_3_FLASH instead.
GEMINI_3_FLASH	Gemini 3 Flash.
GEMINI_3_1_PRO	Gemini 3.1 Pro.

ScreenType Enum

Value	Description
SCREEN_TYPE_UNSPECIFIED	Defaults to DESIGN.
DESIGN	A design screen.
IMAGE	A pure image.
PROTOTYPE	A design prototype.
DOCUMENT	A document (e.g., PRD).
PROTOTYPE_V2	A design prototype v2.

ColorMode Enum

Value	Description
COLOR_MODE_UNSPECIFIED	Unspecified.
LIGHT	Light mode.
DARK	Dark mode.

ColorVariant Enum

Value	Description
COLOR_VARIANT_UNSPECIFIED	Unspecified.
MONOCHROME	Monochrome variant.
NEUTRAL	Neutral variant.
TONAL_SPOT	Tonal spot variant.
VIBRANT	Vibrant variant.
EXPRESSIVE	Expressive variant.
FIDELITY	Fidelity variant.
CONTENT	Content variant.
RAINBOW	Rainbow variant.
FRUIT_SALAD	Fruit salad variant.

Font Enum

29 supported font families:

INTER · DM_SANS · GEIST · SORA · MANROPE · LEXEND · EPILOGUE · BE_VIETNAM_PRO · PLUS_JAKARTA_SANS · PUBLIC_SANS · SPACE_GROTESK · SPLINE_SANS · WORK_SANS · MONTSERRAT · METROPOLIS · SOURCE_SANS_THREE · NUNITO_SANS · ARIMO · HANKEN_GROTESK · RUBIK · IBM_PLEX_SANS · NEWSREADER · NOTO_SERIF · DOMINE · LIBRE_CASLON_TEXT · EB_GARAMOND · LITERATA · SOURCE_SERIF_FOUR

Roundness Enum

Value	Description
ROUNDNESS_UNSPECIFIED	Unspecified.
ROUND_TWO	<i>(Deprecated)</i> Round 2.
ROUND_FOUR	Round 4.
ROUND_EIGHT	Round 8.
ROUND_TWELVE	Round 12.
ROUND_FULL	Fully rounded.

CreativeRange Enum

Value	Description
CREATIVE_RANGE_UNSPECIFIED	Unspecified.
REFINE	Subtle refinements, close to original.
EXPLORE	Balanced exploration. <i>(Default)</i>
REIMAGINE	Radical explorations.

VariantAspect Enum

Value	Description
VARIANT_ASPECT_UNSPECIFIED	Unspecified.
LAYOUT	Element arrangement.

COLOR_SCHEME	Color palette.
IMAGES	Image usage.
TEXT_FONT	Font choices.
TEXT_CONTENT	Text content.

MCP Annotations

MCP Annotations are metadata hints that an MCP server attaches to each tool definition to tell the client (the AI agent or host application) about the tool’s behavioral characteristics. They’re part of the [MCP spec](#) and help clients make smarter decisions about tool usage without needing to understand the tool’s implementation.

Each tool includes behavior annotations:

Annotation	Meaning
readOnlyHint	true = read-only, no mutations.
destructiveHint	true = may create/modify/delete resources.
idempotentHint	true = repeated calls produce same result.
openWorldHint	true = interacts with external systems.

How they work with tools in practice

Each tool in the Stitch MCP server declares these as part of its tool registration. For example:

- `list_projects` has `readOnlyHint: true` — it just fetches data, so an agent can call it without worrying about side effects.

- `create_project` has `readOnlyHint: false`, `destructiveHint: true`, `idempotentHint: false` — it mutates state, so the client should confirm with the user first.
- `generate_screen_from_text` has `destructiveHint: true` and `idempotentHint: false` — each call creates a new screen, so retrying on a timeout could produce duplicates.

Why they matter

These are hints, not enforced rules. They allow MCP clients to:

- Auto-approve read-only tools without prompting the user
- Show confirmation dialogs for destructive operations
- Safely retry idempotent tools on network failures
- Audit external interactions when `openWorldHint` is true

In the Stitch server's case, all tools have `openWorldHint: false` because they interact only with the Stitch backend — not arbitrary third-party services.